# Collective Mpemba cooling beyond triplet tests: a counterexample and a sharp amplitude bound

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**Abstract**

Can tests on every three-state subsystem exclude anomalous cooling of a four-state thermalizing system? We give a strictly positive reversible generator that fails every original-rate triplet test in the proposed finite-temperature protocol, yet exhibits ordinary Mpemba cooling between two finite initial temperatures. Its rational slow eigenpair allows exact verification of the triplet failures and the reversal in relaxation distance. The mechanism is dilution of the slow-mode overlap by a high-energy state whose mode component is nearly zero. A controlled rate family turns this mechanism into a sharp onset criterion. Under the same triplet-failure premise, we prove that the four-state slow-amplitude advantage is strictly less than 25% and construct admissible families approaching this limit. The bound follows from a restriction on the first three components of the slow eigenfunction and the Gibbs mass carried by those states. Lean verifies the rational counterexample, its eventual Pearson chi-square distance reversal, and the sharp amplitude supremum. The result concerns slow amplitudes, not a universal fractional reduction in cooling time.

## 1 Introduction

In Markovian relaxation, a preparation farther from equilibrium can lose its long-lived component more effectively than a closer preparation. The spectral account of the Mpemba effect makes this possibility precise: initial temperature controls the overlap with the slow relaxation mode, while the bath fixes the dynamics [1, 2]. A practical question is whether the absence of an effect in small subsystems can certify its absence in a larger system.

Avitan, Factor and Gelbwaser-Klimovsky proposed an original-rate triplet protocol for this purpose [3]. Its finite-temperature extension infers absence in the full system when every energy-ordered triplet fails a specified necessary-condition test. The inference is a conjecture in that regime. The accompanying zero-temperature theorem uses stronger hypotheses. A later protocol distinguishes exact three-state mechanisms from the earlier necessary-condition predicate [4]; that distinction matters when testing the original conjecture.

We resolve the four-state inference in its regular domain: distinct energies, strictly positive reversible rates, a finite positive bath and a simple slow mode. The inference fails even with strict triplet inequalities and finite cooling preparations. We then explain why it fails and quantify how much ordinary cooling can remain hidden from the tests. The same premise that permits the counterexample enforces a sharp 25% upper bound on the slow-amplitude advantage.

The main text follows these three results: an exact witness, a predictive mechanism and the sharp bound. Figure 1 connects them. The companion *Technical appendices* retain full algebraic proofs, the second counterexample, broader classifications, higher-dimensional extensions and the verification record. The original comprehensive research report is preserved separately. The physical tools used here, including reversible symmetrization, slow-mode projection and Perron theory, are established; the contribution concerns their consequences under this particular triplet premise.

## 2 The model and the precise triplet premise

Use column probability vectors, so $\dot{p} = Mp$ and $a_{ij} = M_{ij}$ is the rate from $j$ to $i$ for $i \neq j$. The diagonal is $M_{jj} = -\sum_{i \neq j} a_{ij}$. Let

$$E_1 < E_2 < E_3 < E_4, \quad q_i = e^{-\beta_b E_i}, \quad Q = \sum_i q_i, \quad \pi_b = q/Q,$$

and impose detailed balance $M_{ij} q_j = M_{ji} q_i$. We assume every off-diagonal rate is positive. Shifting the energy origin and measuring energy in bath units set $E_1 = 0$ and $\beta_b = 1$ without restricting the energy spacings. A common rate multiplier changes only the time unit.

For an ordered triplet $i < j < k$, retain the full system's original rates and define

$$A = a_{ik}, \quad B = a_{jk}, \quad C = a_{ij}, \quad t = q_j/q_i, \quad u = q_k/q_j,$$

$$\begin{aligned} \kappa &= \frac{A+B}{C(1+t)}, \qquad G = At - B, \\ F &= A(1+tu/2) - B(1+u/2) - C(1-t). \end{aligned} \tag{1}$$

The published necessary-condition predicate is

$$P_{ijk} = (\kappa \geq 1 \ \wedge \ F > 0) \ \vee \ (\kappa \leq 1 \ \wedge \ G > 0). \tag{2}$$

Both branch comparisons are inclusive, and both sign comparisons are strict. No effective rates are substituted. If an isolated triplet generator is formed, its diagonals must be recomputed to conserve probability; a principal submatrix is not that generator.

**Lemma 1** (Exact branch reduction)**.** *For positive rates and ordered energies,*

$$\neg P_{ijk} \quad \Longleftrightarrow \quad F_{ijk} \leq 0, \quad G_{ijk} \leq 0.$$

*Proof.* Set $H = A+B-C(1+t)$, $d = (1-t)/(1+t) > 0$ and $h = 2/(1+t)+u/2 > 0$. Direct expansion gives $F = hG + dH$. If $H > 0$, failure means $F \leq 0$, which forces $G < 0$. If $H < 0$, failure means $G \leq 0$, which forces $F < 0$. At $H = 0$, both branches apply and $F = hG$. These cases also prove the converse and include $\kappa = 1$. □

At fixed bath weights the premise is therefore eight closed linear inequalities in the six downward rates, intersected with the positive orthant. Equality faces belong to the original premise, although the individual counterexample below lies strictly inside it.

Let $\lambda_2 < 0$ be the simple slowest nonzero eigenvalue and choose $w^\top M = \lambda_2 w^\top$, $Mv = \lambda_2 v$, $w^\top v = 1$. Then $w^\top \pi_b = 0$, and the slow overlap of a Gibbs preparation is

$$b(\beta) = w^\top \pi(\beta), \qquad \pi_i(\beta) = \frac{e^{-\beta E_i}}{\sum_j e^{-\beta E_j}}. \tag{3}$$

For $0 < \beta_h < \beta_c < 1$, ordinary anomalous cooling means $|b(\beta_h)| < |b(\beta_c)|$ with nonzero overlaps. Define its amplitude advantage by

$$\mathcal{A} = 1 - \frac{|b(\beta_h)|}{|b(\beta_c)|}. \tag{4}$$

The conjectured implication would exclude every such pair whenever all four predicates (2) fail. A single witness in this restricted domain suffices to refute that general finite-temperature inference. It does not challenge the zero-temperature theorem or the three-state necessary condition itself.

## 3 An exact four-state counterexample

**Theorem 1** (Strict triplet failure with ordinary cooling)**.** *There is a positive reversible four-state generator with distinct ordered energies, a finite bath and a simple slow mode for which all four original-rate triplet predicates fail strictly, yet the preparations $\beta_h = 1/1000$ and $\beta_c = 1/100$ exhibit an initial-to-eventual relaxation-distance reversal.*

Take

$$q = (1, 199/200, 2/5, 1/400), \quad E = (0, \log(200/199), \log(5/2), \log 400). \tag{5}$$

The downward rates, in order $(12, 13, 14, 23, 24, 34)$, are

$$\left(1, 28, \frac{249}{10}, \frac{330154675693}{1470375000}, \frac{4983}{200}, \frac{1527327066}{151990625}\right). \tag{6}$$

Set $a_{ji} = a_{ij}q_j/q_i$ for $i < j$ and use conservative diagonals. These formulas specify all entries of $M$ exactly. They immediately give positive rates, detailed balance and column conservation; equilibrium is $\pi_b = (400, 398, 160, 1)/959$.

The following rational bounds summarize exact evaluation of the original predicates. They are sign certificates, not rounded numerical tests.

| Triplet | Branch | Bound on $F$ | Bound on $G$ |
|---|---|---|---|
| 123 | $\kappa > 1$ | $F < -230$ | $G < -196$ |
| 124 | $\kappa > 1$ | $F < -1/50$ | $G < -13/100$ |
| 134 | $\kappa < 1$ | $F < -19/10$ | $G < -2/25$ |
| 234 | $\kappa < 1$ | $F < -118$ | $G < -3/100$ |

All four tests fail, with both branches represented. The exact eigenfunction and eigenvalue are

$$w = \left(1, -\frac{759}{1000}, -\frac{48957}{80000}, -\frac{1}{250}\right), \qquad \lambda_2 = -\frac{4966671}{250000}. \tag{7}$$

Rational multiplication verifies $M^\top w = \lambda_2 w$ and $q^\top w = 0$. The paired right mode $v = \mathrm{diag}(q)w/N$ has $w^\top v = 1$, where

$$N = \sum_i q_i w_i^2 = \frac{27567998009}{16000000000}.$$

The characteristic polynomial factors into $x$, $250000x + 4966671$, and a quadratic with one root in each of $(-337, -336)$ and $(-60, -59)$. Opposite exact endpoint signs locate both quadratic roots; they exhaust its degree. Thus the displayed eigenvalue in $(-20, -19)$ is the simple slow mode, separated from the other nonstationary modes by more than 39. The quadratic and its rational coefficients are printed in the technical appendix on the rational companion.

Integer-power enclosures for $q_i^{1/1000}$ and $q_i^{1/100}$ give

$$\begin{aligned} b(1/1000) &\in (-0.093755177882, -0.093755177881), \\ b(1/100) &\in (-0.093857287853, -0.093857287852). \end{aligned} \tag{8}$$

Both overlaps are nonzero and negative, with the hotter preparation having the smaller absolute value. Every displayed decimal endpoint is a rational number whose enclosing inequalities are checked exactly. Lean independently establishes wider, disjoint intervals sufficient for the same conclusion. The amplitude advantage for this witness is about 0.10879%; its role is to establish strict failure of the implication.

### 3.1 From a slow overlap to an actual distance reversal

For this generator define

$$D_q(p) = \sum_i \frac{(p_i - \pi_{bi})^2}{q_i}. \tag{9}$$

It is a positive multiple of Pearson chi-square divergence and the square of a weighted Euclidean norm. Reversible spectral evolution gives

$$p(t;\beta) - \pi_b = e^{\lambda_2 t} b(\beta) v + O(e^{\lambda_3 t}), \qquad \lambda_3 < \lambda_2. \tag{10}$$

Consequently

$$\lim_{t\to\infty} \frac{D_q(p(t;\beta_h))}{D_q(p(t;\beta_c))} = \frac{b(\beta_h)^2}{b(\beta_c)^2} < 1.$$

For the two specified preparations, exact bounds also give $D_q(\pi(1/100)) < 24 < D_q(\pi(1/1000))$. The hotter preparation therefore starts farther away and becomes closer for every sufficiently large time. Continuity ensures a crossing at a finite positive time. Lean proves the evolution's master equation and uniqueness, its spectral limit and these two distance orderings; the argument does not assume a numerical eigenvector or crossing time.

The conventional KL conclusion follows analytically from

$$D_{\mathrm{KL}}(p(t;\beta)\|\pi_b) = \frac{1}{2} e^{2\lambda_2 t} b(\beta)^2 \sum_i \frac{v_i^2}{\pi_{bi}} + o(e^{2\lambda_2 t}),$$

and $\partial_\beta D_{\mathrm{KL}}(\pi(\beta)\|\pi_b) = (\beta - 1)\,\mathrm{Var}_{\pi(\beta)}(E) < 0$ on cooling. This KL bridge is not part of the Lean verification. Nor does the overlap comparison identify the earliest crossing or imply a reversal at every time. Figure 1A illustrates the quadratic-distance evolution numerically.

## 4 The mechanism: a nearly neutral high-energy state

The slow eigenfunction in Eq. (7) assigns a positive value to the ground state, appreciably negative values to the two middle states and an almost zero value to the highest state. At high initial temperature, population in the highest state dilutes the negative mean overlap. A slightly cooler preparation depopulates that state and increases the magnitude of the mean. Further cooling transfers weight to the ground state and brings the overlap back to zero at the bath. This is a collective projection effect: each isolated triplet has its own eigenfunction and normalization, so its test cannot constrain the full four-state Gibbs projection in the required way.

This explanation can be made predictive by designing a rate family with the desired mode. Keep Eq. (5) and four of the rates, and set

$$\begin{aligned} w(z) &= (1, -p, -r_0 - \eta z, z), \\ p &= 759/1000, \qquad r_0 = 48959/80000, \qquad \eta = 1/160, \\ -3/100 &\le z \le -1/250. \end{aligned} \tag{11}$$

The small adjustment in the third component enforces $q^\top w(z) = 0$. Choose

$$\begin{aligned} a_{23}(z) &= \frac{528238451563 - 2257386450z}{200000(11761 - 500z)}, \\ a_{34}(z) &= \frac{2(1550z^2 - 5989657z + 1197903)}{5(80500z + 48959)}, \end{aligned} \tag{12}$$

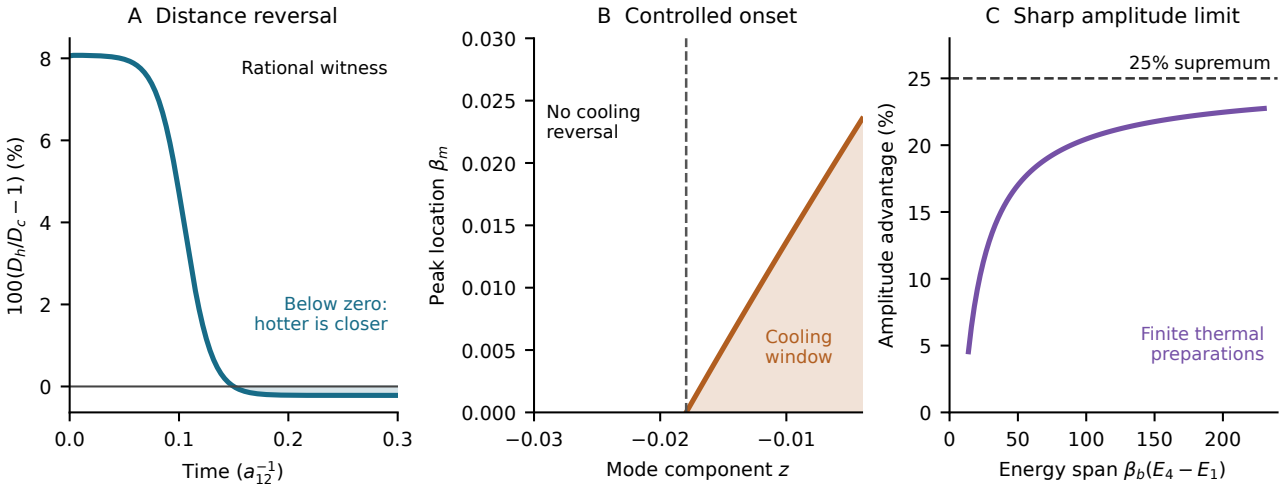



Figure 1: **Counterexample, mechanism and bound. A:** Actual $D_q$ relaxation for the rational generator in Eqs. (5)–(6), with $\beta_h = 1/1000$ and $\beta_c = 1/100$. The sign changes from hotter-farther to hotter-closer; the late-time ratio difference approaches $-0.21747\%$. **B:** Varying the highest-state mode component $z$ in the controlled family produces an anomalous interval $0 < \beta < \beta_m(z)$ only above $z_* \simeq -0.0179401112$. Every triplet still fails throughout this family. **C:** A different family approaches the sharp 25% amplitude supremum as its energy span grows. Here $\beta_h = [\log(1/\varepsilon)]^{-2}$ and $\beta_c$ is the numerical overlap maximum. Curves are high-precision illustrations of separately proved claims. Panel A does not certify a numerical crossing bracket; panel C concerns amplitude, not cooling-time reduction.

while $a_{12} = 1$, $a_{13} = 28$, $a_{14} = 249/10$ and $a_{24} = 4983/200$. Upward rates and diagonals follow as before. Then $M(z)^\top w(z) = \lambda(z) w(z)$ exactly, where $\lambda(z) = -(1550z + 3973343)/200000$. At $z = -1/250$ this is the counterexample in Section 3.

Throughout the closed parameter interval all rates are positive, all four triplets fail strictly, and $w(z)$ is the simple slow mode. These are interval-wide statements: rational Bernstein sign certificates verify the rate and triplet inequalities. A reversible comparison with the baseline rates $a_{23} = 224$, $a_{34} = 10$ places two eigenvalues below $-50$, whereas $-20 < \lambda(z) < -19$. Thus the prescribed vector is not accidentally a fast mode. Full certificates are in the mechanism appendix.

### 4.1 An exact onset criterion

Write $r = r_0 + \eta z$ and $E_\Sigma = E_2 + E_3 + E_4$. In this family $p > r > -z > 0$, $E_4 > E_2 + E_3$, and $(1+r)E_3 > (p-r)(E_3 - E_2)$. The numerator $N(\beta) = \sum_i w_i e^{-\beta E_i}$ has $N'(\beta) > 0$ and $N(1) = 0$, hence $b(\beta) < 0$ on cooling. Put

$$D(\beta) = Z(\beta)^2 b'(\beta), \qquad J(\beta) = e^{\beta E_4} D(\beta).$$

Define $c_{1j} = (w_1 - w_j)E_j > 0$ and $c_{ij} = (w_j - w_i)(E_j - E_i) > 0$ for $2 \le i < j \le 4$. The pair expansion of the covariance gives

$$\begin{aligned} J(\beta) = {} & c_{12} e^{\beta(E_4 - E_2)} + e^{\beta(E_4 - E_2 - E_3)}(c_{13} e^{\beta E_2} - c_{23}) \\ & + c_{14} - c_{24} e^{-\beta E_2} - c_{34} e^{-\beta E_3}. \end{aligned} \tag{13}$$

Because $c_{13} > c_{23}$, every nonconstant term is strictly increasing for $\beta \ge 0$, and $J(1) > 0$. Thus an absolute-overlap maximum occurs on $0 < \beta < 1$ exactly when $J(0) < 0$, and it is unique. This

monotonicity argument is needed: an endpoint derivative alone would not generally exclude another turning point.

Set

$$U = (1 - p - r_0)E_\Sigma + 4(pE_2 + r_0E_3),$$
$$V = (3 + \eta)E_4 - (1 - \eta)E_2 - (1 + 3\eta)E_3 > 0. \tag{14}$$

Direct evaluation yields $J(0) = U - Vz$. Hence

$$\text{ordinary asymptotic cooling exists} \quad \Longleftrightarrow \quad z > z_* := U/V, \tag{15}$$

with the certified bracket $-0.017940112 < z_* < -0.017940111$. For $z \le z_*$, $|b|$ decreases throughout cooling. For $z > z_*$, the interval $0 < \beta < \beta_m(z)$ has increasing $|b|$ and supplies finite-temperature witnesses. Infinite temperature is used only as an analytic endpoint, not as an initial preparation.

The fixed pair $(1/1000, 1/100)$ reverses only for $z > z_{\text{pair}} \simeq -0.01480024798314$. Between $z_*$ and $z_{\text{pair}}$ the effect exists but this pair misses it. Near onset, with $d_* = \partial_\beta D(0, z_*) > 0$ and $\delta = z - z_*$,

$$\beta_m = \frac{V}{d_*}\delta + O(\delta^2), \qquad |b(\beta_m)| - |b(0)| = \frac{V^2}{32 d_*}\delta^2 + O(\delta^3).$$

The anomalous temperature window opens linearly, while its amplitude advantage is initially quadratic. This explains why a finite temperature grid can miss the onset without implying absence.

## 5 Why every such four-state advantage is below 25%

The mechanism suggests a four-to-three-state normalization effect. The following theorem proves that the triplet premise permits no larger ordinary cooling amplitude advantage, regardless of energy spacings or rate ratios.

**Theorem 2** (Sharp amplitude bound)**.** *Let a positive reversible four-state generator have distinct ordered energies, a finite positive bath and a simple slow mode. If all four original predicates (2) fail, then for every $0 < \beta_h < \beta_c < \beta_b$,*

$$\frac{|b(\beta_c)|}{|b(\beta_h)|} < \frac{4}{3}, \qquad \mathcal{A} < \frac{1}{4}. \tag{16}$$

*The value $1/4$ is the supremum of attainable ordinary cooling advantages under these hypotheses and is not attained at finite admissible parameters.*

The proof separates a restriction on the eigenfunction from a thermal normalization inequality. The eigenfunction restriction is the step at which the full $F, G$ premise is essential.

**Lemma 2** (First-three mode restriction)**.** *Under Theorem 2's hypotheses, the slow mode can be oriented so that its overlap is negative on $0 < \beta < \beta_b$ and*

$$w_1 \ge w_2, \qquad w_1 + w_2 \ge 2w_3, \qquad w_4 \le 0. \tag{17}$$

To describe the structural argument, let $Q_k = \sum_{i \le k} q_i$ and use the three bath-orthogonal prefix contrasts

$$h_i^{(k)} = \begin{cases} 1, & i \le k, \\ -Q_k/q_{k+1}, & i = k+1, \\ 0, & i > k+1. \end{cases}$$

The conditions $G_{ijk} \le 0$ imply that the representation of $M^\top$ in the normalized contrast basis is symmetric with nonnegative off-diagonal entries. Its largest eigenvector can be chosen nonnegative. Each contrast has strictly negative Gibbs mean on cooling, so a nonzero nonnegative combination has negative overlap. This also gives $w_1 \ge w_2$ and $w_4 \le 0$.

The remaining inequality, $w_1 + w_2 \ge 2w_3$, uses the $F$ constraints. Subtract the smallest reversible reset component, which leaves $F, G$ unchanged and shifts all nonstationary eigenvalues equally. The residual generator has nonnegative rates and at least one of its first-row downward rates zero. Eliminate the third contrast coordinate by a reversible Schur complement. The desired eigenvector ratio then reduces to a polynomial nonnegativity statement on three exhaustive zero-rate faces, each implied by the original inequalities. Simplicity supplies the strict diagonal gap needed for elimination, including reducible residual generators. The complete face calculation is in the sharp-bound appendix and is formalized in Lean; no sign or mode-shape assumption is added to the theorem's premise.

*Thermal part of Theorem 2.* Normalize $\beta_b = 1$. Let $m(\beta)$ be the conditional Gibbs mean of $w$ on states $1, 2, 3$. Equation (17) expresses those three mode values as a constant vector plus

$$u(1, -1, 0) + v(1, 1, -2), \quad u = (w_1 - w_2)/2 \ge 0, \quad v = (w_1 + w_2 - 2w_3)/6 \ge 0.$$

Both nonconstant vectors have increasing conditional Gibbs mean with inverse temperature. Explicitly, write $a = E_2$, $b = E_3$, $t = e^{-\beta a}$, $y = e^{-\beta b}$ and $(p_1, p_2, p_3) = (1, t, y)/(1 + t + y)$. Then

$$(p_1 - p_2)' = \frac{2at + by - (b - a)ty}{(1 + t + y)^2} > 0, \qquad (p_1 + p_2 - 2p_3)' = -3p_3' > 0.$$

Here $0 < t \le 1$ and $0 < a < b$. Thus $m$ is nondecreasing.

Let $P(\beta) = Z_3(\beta)/Z_4(\beta)$ denote the mass on the first three states. It increases strictly with $\beta$ and obeys $3/4 < P(\beta) < 1$ at finite positive temperature. For $r = P(\beta_c)/P(\beta_h)$, therefore $1 < r < 4/3$. Since $b = Pm + (1 - P)w_4$,

$$b(\beta_c) - rb(\beta_h) = P(\beta_c)[m(\beta_c) - m(\beta_h)] + (1 - r)w_4 \ge 0.$$

Both overlaps are negative, giving $|b(\beta_c)| \le r|b(\beta_h)| < (4/3)|b(\beta_h)|$. This proves the strict bound. The construction below proves sharpness. □

The calculation identifies the physical bottleneck. Depopulating the highest state can amplify the negative overlap of the first three, but their conditional mean cannot become more negative as cooling proceeds. Their probability mass can increase by a factor less than $4/3$ because it already exceeds $3/4$ at every finite positive initial temperature. That thermal restriction turns the local-rate constraints into a global amplitude bound.

## 6 Approaching the bound with admissible generators

Set

$$\begin{aligned} q_\varepsilon &= (1, 1 - \varepsilon/3, 3/5, \varepsilon^2), \\ w_\varepsilon &= (1, -4/5, -1/3 - 4\varepsilon/9 + 5\varepsilon^3/3, -\varepsilon). \end{aligned} \tag{18}$$

The compensation in the third component makes $q_\varepsilon^\top w_\varepsilon = 0$. There are positive reversible generators with this mode and $\lambda_2 = -3$ for every sufficiently small positive $\varepsilon$, and every original triplet predicate

fails. One construction fixes $a_{12} = 1$ and defines the other rates by rational functions. Their limits, in the order used in Eq. (6), are

$$(1, 3/2, 20/17, 15/7, 20/17, 12/17).$$

In this family $F_{124} = G_{234} = 0$ identically. All rates and the remaining required strict signs are controlled by continuity of explicit rational functions at zero; the sign of the vanishing $G_{124}$ factor is checked after extracting its positive factor $\varepsilon$. A positive contrast eigenvector and two connecting positive entries establish slow-mode simplicity. The exact rates and sign checks are in the formal sharpness appendix and `formal/Mpemba/SharpFamily.lean`. The singular bath at $\varepsilon = 0$ is used only to compute limits; every actual member uses $\varepsilon > 0$.

To exhibit finite preparations approaching the bound, take a real parameter $L \to \infty$ and set

$$\varepsilon = e^{-L^2}, \qquad \beta_h = L^{-3}, \qquad \beta_c = L^{-1}.$$

For sufficiently large finite $L$, $0 < \beta_h < \beta_c < 1$. The highest energy is $E_4 = 2L^2$. Consequently its unnormalized Gibbs weight is $e^{-2/L} \to 1$ in the hot preparation and $e^{-2L} \to 0$ in the cooler preparation, while the first three weights tend to one in both. Meanwhile $w_\varepsilon \to (1, -4/5, -1/3, 0)$, and therefore

$$b_h \longrightarrow \frac{1 - 4/5 - 1/3}{4} = -\frac{1}{30}, \qquad b_c \longrightarrow \frac{1 - 4/5 - 1/3}{3} = -\frac{2}{45}.$$

Thus $\mathcal{A} \to 1 - (3/4) = 1/4$. Lean proves that these are limits of actual admissible family members, not merely of a conditional overlap formula, and proves the least-upper-bound and nonattainment statements.

Equality in two triplet constraints is permitted by the original strict predicate tests. A separate construction in the technical appendices has the same limiting weights and eigenfunction, a slow eigenvalue fixed at $-3$, and a gap exceeding $1/3$. A perturbation makes all eight triplet inequalities strict while preserving a gap greater than $1/4$ and the same amplitude limit. Those stronger claims have analytic and exact rational certificates but are outside the present Lean coverage. Figure 1C uses that unperturbed, uniformly gapped family; the thermal geometry is common to both constructions.

The large advantage is accompanied by large energy separation. For example, the separately certified finite choice $\varepsilon = 10^{-50}$, $\beta_h = 1/10000$ and $\beta_c = 1/50$ gives an amplitude advantage between 22.6171597478% and 22.6171597506% before the strict perturbation, and greater than 22% afterwards. This is a finite mathematical witness, not a demonstrated laboratory operating regime.

## 7 Verification, interpretation and scope

The exact rates, eigenpair and triplet signs of Section 3 can be checked using rational arithmetic. Integer-power inequalities certify the thermal enclosures. The companion Lean library goes further: it checks a complete quadratic-distance reversal under the master equation and proves the sharp amplitude theorem for the full generator under explicit spectral assumptions. In the general theorem, `SimpleSlowMode` specifies a nonzero bath-orthogonal left eigenfunction, its eigen-equation, a one-dimensional eigenspace and the largest Rayleigh quotient on the bath-orthogonal subspace.

The main formal entry points are `Mpemba.appendixB_counterexample`, `Mpemba.master_equation_distance_reversal`, and `Mpemba.Quarter.sharp_quarter_isLUB`. The name "Appendix B" refers to the witness's location in the preserved research report; it is the main counterexample of this focused paper. Lean checks the original inclusive branch conditions, exact real Gibbs overlaps, slow-mode simplicity, the evolution and its uniqueness, and the limiting family's admissibility. It does not formalize the controlled onset theorem, the general KL bridge, numerical crossing times, or the uniformly gapped strict-triplet perturbation.

The library pins Lean 4.19.0 and mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. With elan, Git, Python 3 and a POSIX shell installed, preserve the committed dependency manifest and run from the source-package root:

```
cd formal
lake exe cache get
./check.sh
```

The first setup requires network access. Do not run `lake update` to reproduce the pinned dependency set. The README supplies a Lean interpreter fallback if the compiled cache downloader fails to link. The verified build includes 27 proof modules and 43 named-theorem axiom audits, with no admitted proofs or custom axioms. Successful execution ends with

```
PASS: 43 theorem axiom audits; standard logical axioms only.
```

The only permitted axiom dependencies are `propext`, `Classical.choice` and `Quot.sound`. Formal checking does not establish priority or replace review of the correspondence between the source conjecture and its transcription.

The asymptotic distance and time interpretations must be kept separate from amplitude. For a quadratic distance, including the leading KL term, $\mathcal{A} < 1/4$ implies a limiting distance advantage below $1 - (3/4)^2 = 7/16$. For a vanishing common distance threshold, the asymptotic time advance is

$$\Delta t = \frac{\log(|b_c|/|b_h|)}{|\lambda_2|} < \frac{\log(4/3)}{|\lambda_2|}.$$

Both preparations retain the same slow exponent. This is an ordinary effect and does not assert a 25% reduction in total cooling time, which depends on the chosen threshold and earlier dynamics.

The witness also does not contradict finite sampling within a restricted parameter box: its bath-scaled energy span is $\log 400 > 5$. Strict rates, strict triplet signs, spectral separation and strict overlap reversal give an open neighborhood of counterexamples by continuity. No numerical radius for that neighborhood or experimental robustness claim is made.

## 8 Discussion

The failed inference is local-to-global: all original-rate triplet tests can fail while the full system's slow eigenfunction still has anomalous thermal ordering. The nearly neutral high-energy state explains this possibility and supplies a rate-controlled onset. The full triplet premise nonetheless constrains the slow eigenfunction strongly enough to bound the four-state amplitude advantage sharply.

The distinction between the counterexample and the bound is useful. The former disproves a universal absence claim with an interior witness; the latter describes the entire regular four-state domain under that premise. The limiting factor 4/3 has a direct Gibbs-normalization interpretation, and the matching construction shows that it cannot be improved without additional assumptions.

The technical appendices retain the broader results: surviving strong-effect and heating exclusions, finite-temperature classifiers, degenerate-mode results, finite-time distance tests, inverse realization, finite-span refinements and higher-dimensional constructions. These are not needed to state the central four-state conclusion. A useful further physical question is how much finite-time advantage remains when energy spans and rates are constrained by a concrete thermalization model.

**Availability.** The accompanying source package contains this manuscript, the technical appendices, the preserved comprehensive research report, exact Python certificates, the Lean library and the overview figure's source and data. `RUNME.txt` gives the document and verification build commands. The figure is generated by `research/focused_overview.py`; its curves are numerical illustrations, with exact claims supported by the separate proofs and certificates described above.

## References


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